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INDUSTRIES



FOREX DSS

Enterprise Computing Phase 2

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Stage 1:

1.1. Stakeholder Identification and Analysis

The Forex Trading Decision Support System developed in Phase 1 serves multiple distinct stakeholder groups, each with different requirements for data visualisation and decision making support. Identifying these stakeholders is essential to understanding how data must be presented to meet their specific needs and operational contexts. The primary stakeholder groups for this DSS are novice to intermediate Forex traders, trading coaches and mentors, institutional fund managers, and system administrators. Each group requires fundamentally different visualisations because they operate at different levels of decision making granularity and possess different levels of technical trading knowledge.

1.2. Analysis of Stakeholder Requirements

Novice to Intermediate Traders (Primary End Users)

Novice and intermediate Forex traders represent the primary stakeholders for whom the DSS was originally designed. These users require data visualisations that translate complex technical analysis into clear, actionable trading signals without requiring deep understanding of the underlying mathematical models. Their primary need is immediate signal recognition and rapid decision making under time pressure, since currency markets move continuously during trading sessions. These traders must quickly understand entry price, stop loss level, take profit target, and confidence score without requiring extensive explanation or technical context.

The data story for this stakeholder is straightforward: “take this trade now, or wait.” The trend they must interpret is the immediate direction of the signal, the pattern is whether the current setup resembles past profitable setups, and the relationship is between the confidence score and the risk-reward tier offered. The visualisation purpose for this stakeholder group is to reduce cognitive load by presenting

signal information in a format that requires minimal interpretation, enabling traders to execute recommendations with confidence within seconds of signal generation.

Trading Coaches and Mentors (Educational Stakeholders)

Trading coaches who work with novice traders represent a secondary but highly important stakeholder group. These users require visualisations that expose the reasoning behind each signal so they can teach their mentees why a particular setup is valid and how the system evaluated market conditions.

The data story for this stakeholder is “observe how the system reached this decision.” The trend they interpret is how filter outcomes change across different market conditions, the pattern is which combinations of filters tend to precede profitable trades, and the relationship is between displacement strength and trade outcome. Coaches need visibility into the reasoning pipeline including which structural events triggered, how the displacement ratio (the size of the breakout candle body compared to the recent average body) was calculated, and which market condition filters blocked invalid signals. The visualisation purpose here shifts from rapid decision-making to transparency and education.

Institutional Fund Managers and Risk Officers (Governance Stakeholders)

Institutional fund managers and compliance officers occupy a third stakeholder group with distinct visualisation requirements. These users are concerned with portfolio-level aggregate statistics, cumulative profitability, risk exposure, win-rate consistency, and drawdown management across multiple currency pairs and time horizons. They do not require individual signal-level detail; instead, they require summary dashboards showing whether the DSS is performing within risk tolerance bands.

The data story for institutional stakeholders is “is this system preserving capital and delivering acceptable returns within our risk parameters?” The translation they interpret is the cumulative equity curve over weeks and months, the pattern is the consistency of win rate across pairs, and the relationship is between risk taken (drawdown) and reward earned (return). The visualisation purpose is high-level governance and reporting to compliance frameworks.

System Administrators and Technical Analysts (Operational Stakeholders)

System administrators and technical analysts who monitor DSS health represent a final stakeholder group. These users require visualisations focused on system operability, error rates, data freshness, processing latency, and market data quality. Their data story centres on system reliability, “is the indicator executing correctly on all configured pairs, are real time data feeds maintaining integrity, are there any edge cases or anomalies requiring attention?” This stakeholder requires different visualisation altogether, focused on operational metrics rather than trading outcomes.

1.3. Explanation of How Stakeholders Use DSS Outputs to Support Decision Making

Trader Level Decision Making

System administrators and technical analysts who monitor DSS health represent a final stakeholder group. These users require visualisations focused on system operability, error rates, data freshness, and data quality.

The data story for this stakeholder is “is the system functioning correctly so that the results can be trusted?” The trend they interpret is whether error rates are rising over successive evaluation cycles, the pattern is whether failures cluster around particular data conditions, and the relationship is between data completeness and successful signal generation. This stakeholder requires operational metrics rather than trading outcomes.

Coach Level Decision Making

Trading coaches use DSS outputs to make educational decisions about what to teach their students. When reviewing a profitable trade, a coach uses the detailed signal explanation and filter status visualisation to build a teaching narrative, “notice how the market structure shift candle was 2.3 times the average body size, which gave us a confidence score of 0.75 for a 2:1 risk reward ratio , this is why the take profit target was set aggressively rather than conservatively.” Conversely, when reviewing a rejected setup, the coach uses the filter rejection explanation to teach students about market condition quality, “here you see the regime filter blocked this signal because the market was in consolidation mode, even though the structural setup looked valid , this is how the system protects you from high probability false signals in ranging conditions.” The coach focused visualisation must therefore preserve detailed reasoning information and make filter evaluation criteria visible and interpretable.

Institutional Decision Making

Institutional fund managers use aggregate DSS outputs to make strategic capital allocation decisions. At the end of each week or month, they review cumulative win rate, average win size compared to average loss size, maximum drawdown experienced, Sharpe ratio, and whether the DSS is operating within pre established risk tolerance bands. If the DSS shows a 65% win rate with a 2:1 average risk reward, the manager decides whether to allocate additional capital to the strategy. If a drawdown exceeds 8% of allocated capital, the manager may decide to reduce position sizing or temporarily pause the system. These are portfolio level decisions that occur at weekly or monthly intervals, not in real time like trader decisions. The institutional visualisation must aggregate and summarize hundreds of individual trades into performance metrics that institutional compliance frameworks understand.

Operational Decision Making

System administrators use technical health metrics to make operational decisions about system maintenance and troubleshooting. If the visualisation shows that real time data latency has increased from 50 milliseconds to 2 seconds, the administrator investigates the data feed connection. If error rates on a particular currency pair spike above historical baseline, the administrator checks for data anomalies or temporary market disruptions. If the system stops executing signals altogether, the administrator checks broker connection status and licensing. These visualisations support diagnosis and remediation decisions rather than trading or strategy decisions.

1.4. Analysis of Data Integrity Considerations

Data Ownership and Source Reliability

The Forex DSS combines multiple data sources, and understanding the ownership and reliability of each is essential for stakeholder trust. In the Phase 2 Excel implementation, The historical OHLC price data is imported once from a broker or data provider and stored as a static dataset within the workbook, rather than streamed live. This OHLC data originates from regulated trading venues and is highly reliable because it represents actual executed trades. Ownership of this data belongs to the broker or venue that recorded it, and it is used here under the provider's permitted terms for analysis.

Derived indicator data (ATR, EMA, and the displacement and efficiency measures) is calculated internally by the DSS through formulas applied to the stored OHLC data. Its reliability is high because the calculation logic is deterministic and repeatable, but it remains dependent on the quality of the imported price data. If the imported data contains errors, every derived value inherits those errors.

Because the dataset is static, stakeholders must understand that all insights reflect the market conditions captured at the time of import, not the live market. This is an acceptable and deliberate constraint for a planning and decision-support tool, but it means the data's age must always be communicated clearly to prevent misinterpretation.

Data Validation Processes

The DSS implements validation at multiple stages. At input, the worksheet enforces constraints on what the user may enter, so dates must be valid calendar dates and prices must fall within a sensible range for a currency pair. This prevents nonsensical inputs from corrupting downstream calculations. At calculation, division operations are protected by a small non-zero floor so that an empty or malformed cell cannot produce an invalid result. At decision, the system confirmed that all eight filter conditions have been evaluated before a take-or-reject result is issued, so that no signal is produced from incomplete data.

Assessment of Risks of Incorrect or Misleading Interpretation

Overconfidence Bias from High Confidence Scoring

A primary risk is that traders place excessive confidence in high confidence scores. The confidence score reflects displacement strength only (how explosively price moved during the market structure shift candle), not the probability of profit. A high-confidence signal can still fail in unfavourable conditions. To mitigate this, trader visualisations define the confidence score as displacement-strength-based rather than a probability of profit, and coach visualisations reinforce that position sizing must always respect the risk-reward ratio regardless of confidence.

Misinterpretation of Rejection Reasons

A second risk is that traders misinterpret why a signal was rejected and conclude the system is broken. If the regime filter blocks signals during consolidation, an unfamiliar trader might believe the DSS has failed when it is in fact protecting against false signals. To mitigate this, filter rejection is always accompanied by a plain-language explanation stating which condition failed and why, so the user understands that rejection is protective.

Data Freshness and Stale Data Interpretation

A critical risk is the misinterpretation of static data as live performance. Because the Phase 2 system evaluates against a stored historical dataset rather than a live feed, every dashboard must state the date range of the underlying data clearly, for example “data imported for the period ending 15 April 2026.” This prevents a stakeholder from assuming the figures reflect current market conditions. Rather than promising automatic real-time refresh, which Excel cannot provide, the system is honest about its static nature and relies on a clearly labelled manual re-import when newer data is required.

Stage 2:

2.1 Stakeholder Identification and Analysis

[Content copied directly from Phase 1 - Section: Stakeholder Groups]

The intended users of this system are new to intermediate Forex traders who possess basic knowledge of currency markets but lack the experience required to consistently apply technical analysis during live trading conditions [<https://www.investopedia.com/articles/forex/11/why-trade-forex.asp>]. Beyond the primary trader stakeholder, four additional groups represent secondary but important stakeholders. Trading coaches and mentors work with novice traders and require visualisations exposing the reasoning behind each signal for teaching purposes [<https://www.tradingview.com/education/>]. Institutional fund managers and compliance officers require portfolio-level aggregate statistics and risk monitoring dashboards [<https://www.bloomberg.com/professional/product/terminal/>]. System administrators require operational health monitoring dashboards to track whether data feeds are connected and whether the system is functioning correctly [<https://docs.libreoffice.org/>]. Data analysts may require access to raw trade data and historical performance records for research and optimisation purposes [<https://www.investopedia.com/>].

2.2 Research into Similar Trading Dashboards and Financial Visualisation Examples

Because Phase 2 must be delivered entirely in Excel rather than executable Pine Script, research focused on real, published projects that solve the same problem this DSS solves, namely automated signal logic, trade evaluation, and performance reporting built using only spreadsheet formulas and macros instead of a coded indicator. Four projects were selected, each demonstrating an architectural approach that aligns with one of the four stakeholder dashboards.

The diagram below maps each researched project to the stakeholder dashboard it most closely resembles.



Project 1. Piyush Nawade's Excel Backtest Engine (trader-type approach)

Nawade built a fully modular backtesting engine in Excel that generates entry/exit signals from an SMA crossover rule, then auto-logs every trade. The relevant design lesson is that he avoided hardcoding entirely: a single Inputs sheet holds every parameter, and OFFSET and INDEX formulas reference those cells dynamically so changing one value updates the whole workbook. Signal logic lives in a Strategy sheet, with a VBA macro detecting each buy/sell pair and building the trade log automatically. This mirrors the trader dashboard requirement of fast, clean signal output where the user changes inputs and immediately sees the resulting decision without manual recalculation [[An Automated Trading System Built in Excel \(That Actually Tracks Trades, Costs, and Drawdowns\) | by Piyush Nawade | Medium](#)]

The principal strength of this project is the clarity of its single-screen output. Because every parameter lives on one Inputs sheet and every downstream cell references it, the user never has to hunt across the workbook to understand how a result was produced, and the trade log builds itself the instant a new crossover is detected. For a trader stakeholder who must reach a decision quickly, this is precisely the right priority: the cognitive load of interpreting the system is minimised, and the structured trade log doubles as an audit trail that explains what the system did and when.

Its limitation, judged against the needs of this DSS, is that the entire detection step depends on a VBA macro that the user cannot see or interrogate. The crossover logic executes inside code, so a novice cannot inspect why a particular bar did or did not generate a signal, only that the trade log changed.

The single crossover rule is also far simpler than this DSS requires, evaluating one condition rather than eight sequential filters. These limitations directly inform two design decisions. First, this DSS replaces hidden macro detection with visible IF(AND()) formula gates, so the take-or-reject decision can be read and checked in the cells themselves rather than trusted as the output of code. Second, it preserves Nawade's dynamic-referencing discipline, holding all parameters in one place so that the trader dashboard remains as immediately legible as his, while extending the evaluation from one rule to the full eight-filter pipeline.

Project 2. Ahaan28's Momentum-Adjusted Trend Reversal Strategy (coach-type approach)

This GitHub project uses Excel VBA and macros to automate signal generation and build an interactive dashboard that lets the user switch between price data, returns, average returns, and performance charts. Its value as research is the emphasis on transparency: rather than hiding logic, it exposes the intermediate datasets so the user can inspect why the strategy behaved as it did across the backtest period. This is exactly the coach dashboard's purpose, making the reasoning behind each signal visible and switchable for teaching, rather than only showing the final answer [[GitHub - ahaan28/Momentum-Adjusted-Trend-Reversal-Trading-Strategy-using-Excel-VBA-and-Macros](https://github.com/ahaan28/Momentum-Adjusted-Trend-Reversal-Trading-Strategy-using-Excel-VBA-and-Macros): This project highlights how Excel's native automation tools can be leveraged to build practical trading models and backtest their feasibility. The results underline the promise and limitations of simple trend reversal approaches in volatile markets · GitHub]

The principal strength of this project is that it treats the intermediate calculation as something worth showing rather than hiding. By letting the user switch the same dashboard between raw price, period returns, and averaged returns, it lets a viewer trace a result back through the steps that produced it, which is the single most important property a teaching tool can have. A coach using a dashboard built on this principle can move a student from the headline outcome to the underlying numbers without leaving the screen, demonstrating cause and effect rather than asserting it.

Its limitation, against the needs of this DSS, is that the transparency stops at the dataset level rather than the decision level. The user can switch between views of the data, but the project does not present a per-condition pass-or-fail breakdown explaining why a specific signal was accepted or rejected, and its momentum rule again resolves to a single condition rather than a structured set of filters. This informs the coach dashboard design directly: this DSS keeps ahaan28's switchable, inspectable datasets but adds an explicit filter-evaluation table in which each of the eight conditions is shown with its tested value, its pass-or-fail result, and a plain-language explanation. The reasoning is therefore exposed not only at the level of the data, as in ahaan28's project, but at the level of the decision, which is what a coach needs to teach the logic of the system.

Project 3. Chasmani's FOREXCalcExcel (institutional-type approach)

This is the most directly comparable project because it is an Excel-based simulation built specifically for Forex backtesting, the same market this DSS targets. It demonstrates how an entire Forex strategy can be simulated across historical data within Calc and reduced to aggregate performance outputs. This aligns with the institutional dashboard, which needs many individual trades summarised into portfolio-level metrics such as cumulative return, win rate, and drawdown rather than real-time signals [[GitHub - chasmani/FOREXCalcExcel: Excel based simulation for FOREX trading backtesting · GitHub](#)]

The principal strength of this project is its proof that a complete Forex strategy can be run across a body of historical data inside a spreadsheet and collapsed into a small set of summary figures. This is exactly the operation an institutional stakeholder needs, because a fund manager does not read individual trades but asks whether the strategy as a whole preserved capital and produced acceptable returns. By performing the simulation against stored historical data rather than a live feed, it also validates the static-dataset approach this DSS is constrained to use in Excel, confirming that the constraint does not prevent meaningful aggregate analysis.

Its limitation is that it presents a single, undifferentiated set of outputs with no stakeholder-specific framing and no risk-adjusted context. A raw cumulative return figure does not tell a fund manager whether that return was earned safely, and the project offers no equivalent of a Sharpe ratio or a drawdown band against which performance can be judged. This informs the institutional dashboard design in two ways. First, this DSS keeps chasmani's aggregate-from-historical-data method but separates the institutional view from the trader and coach views, so each stakeholder sees only the level of detail relevant to their decision. Second, it adds risk-adjusted and comparative context, presenting win rate, average win against average loss, maximum drawdown and a risk-adjusted return measure alongside the headline figure, so that the aggregate output supports a genuine governance judgement rather than a bare profit number.

Project 4. Samar2598's Stock Trading Analysis Dashboard (operational-type approach)

This MS-Excel dashboard requires the user to enter only a small set of input fields (trade number, stock name, market cap, entry price, stop loss, win/loss flag), and every other column is then calculated automatically from those inputs. Its lesson is the strict separation of manually-entered source data from automatically-derived fields, with validation built into what the user is allowed to enter. This aligns with the operational dashboard, which depends on clean input integrity and automatic status calculation to monitor whether the system is functioning correctly [[GitHub - samar2598/Stock-Trading-Analysis-Dashboard: This is a dashboard created using MS-Excel to analyse the stock related tasks\(eg. trading, back testing strategies etc\) and create visual and interactive interface. · GitHub](#)]

The principal strength of this project is the discipline of its input boundary. By restricting the user to a handful of source fields and deriving everything else by formula, it guarantees that the

manually-entered data is the only place an error can originate, and it makes that data easy to validate at the point of entry. This is the foundation of data integrity in any spreadsheet system, because once the source fields are trustworthy, every derived figure inherits that reliability. For an operational stakeholder whose entire concern is whether the system can be trusted, this clean separation is the most valuable property the project demonstrates.

Its limitation is that it monitors the trade data but not the health of the system that processes it. The dashboard validates what the user types, yet offers no indication of whether the underlying dataset is complete, whether the last calculation cycle succeeded, or how often inputs have produced errors. This informs the operational dashboard design directly. This DSS adopts samar2598's strict input and output separation and point-of-entry validation, then extends it with a dedicated status worksheet that reports the completeness of the historical dataset, the success of the most recent evaluation cycle, and the running error rate. The operational view therefore monitors not only the integrity of the data, as samar2598's project does, but the integrity of the evaluation process itself, which is what a system administrator must be able to confirm before trusting the results.

2.3 Analysis of Strengths and Limitations

Strengths identified across the four projects

All four projects confirm that genuine decision-support logic can be reconstructed in a spreadsheet without code execution. Nawade's dynamic referencing proves that a no-hardcoding architecture keeps a workbook flexible. Ahaan28's switchable datasets prove that filter reasoning can be made transparent for an educational user. FOREXCalcExcel proves the approach scales to Forex specifically and to aggregate reporting. Samar2598's input-then-auto-calculate model proves that strict input/output separation with validation keeps data reliable. Together they validate the core Phase 2 design of eight filter columns resolving to TRUE/FALSE, gated by an IF(AND()) formula or SQL WHERE clause, feeding a take-or-reject output.

Limitations identified across the four projects

The same four projects share limitations this DSS must design around. None integrates a live data feed, as every one relies on manual data entry or periodic import, so signals are evaluated on a delay rather than in real time. Their charts are static images that cannot be clicked to drill down. VBA-dependent projects require macro-enabled files and programming knowledge that an end user may not have, which is a reason this DSS favours native formulas and SQL queries over heavy VBA where possible. Larger trade histories also slow recalculation. The Phase 2 design responds by adopting manual update cycles, native-formula logic gates, and carefully chosen summary tables rather than attempting to replicate real-time interactivity.

2.4 Synthesis of Research Findings

The research into the four comparable Excel-based DSS projects produces a consistent set of design principles that directly inform the planning of this DSS. Each project independently arrives at a three-tier architecture separating raw input data, calculated logic, and visualised output across distinct worksheets, confirming that this separation is the established convention rather than an arbitrary choice.

The planned architecture for Excel must account for a critical constraint: Excel cannot integrate live data feeds from brokers. Instead, the architecture stores historical OHLC data in a hidden read-only worksheet. The user manually imports prior bars of EUR/USD data into this hidden sheet, and when they manually enter a new bar's values into the Input worksheet, formulas evaluate that new bar against the historical dataset to determine filter pass or fail status. This manual-trigger evaluation cycle replaces the real-time autonomous execution that Phase 1 Pine Script provided.

The table below maps each limitation found in the researched projects to the specific design decision it informs in this DSS, following the synthesis format used in Phase 1.

Research Finding (Limitation in existing project)	Application to This DSS (Solution)
Nawade's engine relies on a coded VBA macro to detect signals, requiring programming literacy.	This DSS replaces macro detection with native IF(AND()) formula gates so the take-or-reject decision is readable without code.
ahaan28's dashboard exposes returns but does not explain why each filter passed or failed.	The coach dashboard adds a plain-language explanation column for every filter, making the reasoning fully transparent for teaching.
FOREXCalcExcel aggregates results but offers no stakeholder-specific view of the data.	This DSS separates aggregate institutional reporting from the trader and coach views so each stakeholder sees only what they need.
samar2598's dashboard has no system-health or data-integrity monitoring.	The operational dashboard adds a status worksheet tracking data completeness, query success and error rate.
None of the four researched projects integrates a live data feed.	This DSS is designed around a static historical dataset and a manual trigger cycle, matching the genuine constraints of Excel.

2.5 Explanation of UX/UI Design Principles Applied to the Excel Implementation

The research into existing dashboards reveals five user-experience principles that govern effective spreadsheet design, each applied deliberately across the four DSS dashboards.

Progressive Disclosure

Progressive disclosure recommends presenting essential information immediately while keeping detailed data accessible on demand. Each dashboard's primary worksheet shows only the single most important output, and granular data is relocated to secondary worksheets reached through tabs and hyperlinks.

This principle is applied differently for each stakeholder according to how much detail their decision requires. The trader sees only the take-or-reject signal and a short recent-trades table, with the full history one tab away, because their decision is immediate and detail would slow it. The coach sees the filter table on the primary view but the underlying formula breakdown only on a linked sheet, so a lesson can begin with the result and descend into the calculator when the moment calls for it. The institutional view shows headline metrics first and the trade-by-trade record last, and the operational view shows status indicators first and detailed logs on demand. In every case the most decision-relevant figure is never hidden, and the supporting depth is always one deliberate step away rather than absent.

Consistency

Consistency recommends that visual language and interaction patterns remain uniform so users develop transferable mental models. Column order, table structure, and the green and red colour convention are standardised across every worksheet so a user who learns one dashboard can navigate the others.

The value of this for a multi-stakeholder system is that knowledge transfers between roles. A coach who also trades, or a fund manager who reviews a coach's screen, meets the same colour meanings, the same column order, and the same table layout on every dashboard, so no relearning is required when moving between them. The same date format, the same green-for-positive convention, and the same placement of the headline figure at the top-left recur everywhere, which turns the four dashboards into one coherent system rather than four separate tools that happen to share data.

Visual Hierarchy

Visual hierarchy recommends using font size, weight, and spacing to direct attention to the most important information first. The headline output of each dashboard appears in the largest, boldest type at the top of the worksheet, with supporting detail in progressively smaller type below.

Hierarchy is tuned to each stakeholder's single most important question. On the trader dashboard the largest element is the direction and price of the signal, because that is what a decision turns on. On the coach dashboard the filter table is given the central visual weight, because the teaching happens there. On the institutional dashboard the cumulative return occupies the largest type, because that is the first governance question. On the operational dashboard the status indicators are the dominant element, because an administrator needs to confirm soundness before reading anything else. The eye is therefore guided to the right starting point on each screen rather than to a generic one.

Minimalism

Minimalism recommends removing any column, row, or decorative element that does not convey meaning. Each primary view is limited to the essential fields and defers derived metrics to detail worksheets, following samar2598's model.

Minimalism is most important on the trader and operational dashboards, where speed of reading is the whole point, so these views carry the fewest possible elements and no decorative formatting that does not encode data. It is applied more moderately on the coach dashboard, where additional columns such as the explanation text genuinely carry meaning and so are retained. The test applied throughout is whether an element changes a decision or a lesson, and if it does not, it is moved to a detail sheet or removed, which keeps each primary view free of the clutter that would otherwise compete with the information that matters.

Feedback

Feedback recommends that the system communicate the result of a user action immediately. Data validation messages and conditional formatting respond the instant data is entered, recolouring a profit or loss cell green or red without further action.

Because the system runs on a manual trigger rather than a live feed, immediate feedback is what confirms to the user that their action has taken effect. When a trader enters a new bar, the signal cell recolours and the filter results update at once, confirming the evaluation has run. When an invalid value is typed, a validation message appears at the point of entry rather than allowing a silent error to surface later in a calculation. This responsiveness is the substitute, in a static-data system, for the reassurance that a live system gives through constant updating, and it is the reason the user can trust that what they see reflects what they last entered.

2.6 Description of Input Datasets and Data Architecture for Each Dashboard

The DSS architecture in Excel requires a hidden historical data worksheet and three visible worksheets per stakeholder. All operate on a manual trigger cycle where the user enters a new bar's data and the system evaluates it against the stored historical dataset.

Hidden Historical Data Worksheet

A hidden, read-only worksheet stores 100 to 250 bars of prior EUR/USD OHLC data imported from a CSV file. This dataset remains static unless the user explicitly refreshes it. SQL queries and formulas reference this historical data to calculate baseline metrics such as prior range, recent range, average body size, and volatility bands. This hidden sheet is the source of truth for all comparative calculations, allowing the system to evaluate new bars against established market conditions rather than against real-time data.

Trader Dashboard Data Flow

The trader dashboard draws on three active worksheets. An Input worksheet receives the manually entered new bar data (date, time, open, high, low, close). Upon entry, a VBA macro or formula chain automatically triggers SQL queries that evaluate all eight filter conditions against the historical dataset. A Logic worksheet calculates the confidence score based on displacement strength and resolves the IF(AND()) gate to produce a take or reject decision. A Summary worksheet displays the current signal in large text along with a table of recent trades, all updated after the trigger evaluation completes.

Coach Dashboard Data Flow

The coach dashboard adds a Signal Analysis worksheet that displays the current bar's filter evaluation in full. SQL queries populate the columns showing each filter's condition, the calculated values from the historical data, the pass/fail result, and a plain-language explanation. A secondary Calculation Breakdown worksheet exposes the SQL WHERE clauses and formula logic so coaches can show students exactly how displacement ratio was calculated from the historical dataset and how that ratio fed into the confidence score.

Institutional Dashboard Data Flow

The institutional dashboard draws from a complete Trade History worksheet recording every evaluation made by the user (manually entered bar plus resulting signal). SQL queries aggregate this history using SUMIF and COUNTIF to calculate cumulative returns, win rates, and monthly performance. A pivot table groups trades by symbol and calculates metrics per pair. All institutional metrics are calculated from the accumulating trade history stored in the workbook, not from live market data.

Operational Dashboard Data Flow

The operational dashboard requires a single Status worksheet where system administrators manually log or import status checks. A row for Data Feed completeness shows whether the historical dataset has the required 100+ bars. A row for SQL Validation logs whether queries executed successfully on the last trigger cycle. A row for Error Rate tracks how many signal evaluations failed due to missing data or formula errors. All status information is manually entered or populated by error-handling formulas after each evaluation cycle.

2.7 Justification of Chart and Visualisation Selections

Trader Dashboard Visualisations

A column chart is selected to display the profit and loss of recent trades because it communicates the ratio of winning to losing trades at a glance, with profits rising green above the zero line and losses falling red below it. A line chart is selected for the cumulative equity curve because it reveals at once whether the account is in drawdown or recovery, which a table of values cannot convey as immediately.

A column chart is preferred over a pie chart for the recent-trade results because a pie chart would show only the proportion of wins to losses and would lose the sequence and magnitude of each trade, both of which a trader needs to read a streak. A line chart is preferred over a column chart for the equity curve because the continuous line emphasises the direction and slope of cumulative performance, whereas separate columns would invite the eye to compare individual heights rather than follow the overall trajectory, which is the very thing a trader is trying to judge.

Coach Dashboard Visualisations

A colour-coded filter table is selected as the primary visualisation because the coach's data story depends on tracing pass and fail logic filter by filter, which a chart cannot express. A scatter plot of displacement ratio against trade outcome is selected to demonstrate the relationship between setup strength and profitability.

A table is preferred over any chart for the filter evaluation because the information is categorical and per-condition: each of the eight filters has a name, a tested value, a result and an explanation, and only a table can hold those four fields side by side for every filter at once. A scatter plot is preferred over a line or bar chart for the displacement-against-outcome relationship because the data is a cloud of independent trades rather than an ordered series, and a scatter plot is the only common chart type that shows the relationship between two variables across many unconnected points and lets the coach show that the relationship is a tendency rather than a guarantee.

Institutional Dashboard Visualisations

Key performance indicators are selected as large numeric values rather than charts because institutional stakeholders require headline metrics readable in seconds. A line chart of cumulative return reveals the performance trend, a horizontal bar chart of return by symbol exposes which pairs perform well, and a pivot table permits sorting and filtering across the full trade population.

Large numeric indicators are preferred over charts for the headline figures because a single decisive number such as total return or maximum drawdown is read faster as text than as a shape, and precision matters at the governance level. A horizontal bar chart is preferred over a vertical one for the by-symbol comparison because the currency-pair labels are text that sits more legibly at the left of horizontal bars than beneath crowded vertical columns, and the ranking from best to worst pair is read naturally down the page. A pivot table is preferred over a fixed table for the trade population because it lets the manager re-sort and re-group the same data to answer questions as they arise, without the dashboard having to anticipate every query in advance.

Operational Dashboard Visualisations

Traffic-light status indicators are selected because operational monitoring requires an instant judgement of whether all systems are sound, and sparklines are selected to embed a trend within each status row so an administrator can distinguish a chronic problem from an intermittent one.

A traffic-light indicator is preferred over a numeric readout for status because an administrator scanning for problems needs a categorical operational, degraded or failed judgement rather than a raw figure to interpret, and colour with a label delivers that judgement instantly. A sparkline is preferred over a full chart for the trend because it must fit inside a single status row beside the current value, because a full-sized chart would break the one-row-per-component layout that makes the dashboard scannable, whereas the sparkline conveys just enough shape to separate a steady problem from an intermittent spike without leaving the row.

2.8 Justification of Colour Schemes

Colour is justified as a deliberate design decision rather than decoration, because the marking criteria require the colour scheme to be reasoned against stakeholder needs. Across every dashboard the system uses a single consistent convention: green denotes a positive or passing state (a profitable trade, a filter that passed, a long signal), red denotes a negative or failing state (a losing trade, a filter that failed, a short signal), and a neutral grey denotes an inactive or not-applicable state. This convention is chosen because it matches the near-universal financial-market association of green with gain and red with loss, so traders interpret it without instruction.

The colour scheme is applied through conditional formatting so that meaning is generated by the data itself rather than entered manually, which removes the risk of a cell being mislabelled. On the trader dashboard the headline signal cell is shaded green or red by direction to support a decision in seconds. On the coach dashboard the filter status column is shaded per row so a student can see at a glance how many conditions passed. On the institutional dashboard cells are shaded against tolerance bands, so a drawdown beyond the agreed limit turns red as a warning. On the operational dashboard the green, amber and red traffic-light convention signals operational, degraded and failed states.

Because colour alone disadvantages users with colour vision deficiency, every coloured state is paired with a text label or symbol, so the scheme communicates through both hue and word. This dual encoding is the reason the colour decision is justified rather than merely stylistic, and it is examined further under accessibility in section 2.11.

2.9 Justification of Layout and Placement on Dashboards

Layout and placement are justified against the reading behaviour of each stakeholder, because the marking criteria require placement decisions to be reasoned rather than incidental. The governing principle is that the eye reads a worksheet from the top-left, so the single most important output for each stakeholder is placed in the top-left region where it is seen first.

On the trader dashboard the current take-or-reject signal is placed at the very top in the largest type, with the recent-trades table beneath it and the charts below that, so the decision is reached before any supporting detail competes for attention. On the coach dashboard the price context sits at the top, the eight-filter evaluation table occupies the central focus where teaching attention is directed, and the resulting trade outcome sits beneath it, so the layout physically follows the teaching narrative from observation to reasoning to result. On the institutional dashboard the headline performance indicators are placed across the top as a summary band, the period-comparison table sits immediately below for context, and the granular trade history is placed last because it is consulted only on demand. On the operational dashboard the status indicators are arranged as a single scannable column so that one downward glance confirms whether every component is sound.

Placement also separates input from output. The hidden historical data and the input-entry area are kept away from the visual output region so that a user cannot accidentally overtype a calculated cell, which protects data integrity and follows the strict input/output separation observed in the samar2598 project.

WireFrames

The planned layout for each dashboard is set out in the annotated wireframes below. Each wireframe fixes the position of every zone before the build and maps that position to the UX principle it serves, so the placement decisions described in this section are shown as well as stated. The numbered annotations on each wireframe correspond to the design rationale panel beside it.

wf

The Trader dashboard wireframe places the take or reject decision in the top left, which is where the eye lands first and therefore where the single most important output belongs under the visual hierarchy principle. The six action parameters are grouped immediately beside the decision so related information sits together by proximity, the recent signals table sits below as a scannable record of form, and the two charts anchor the foot of the view so the trader reads the decision before the supporting trend.

wf

The Coach dashboard wireframe gives the eight filter evaluation table the central position because the coach's whole task is tracing pass and fail logic filter by filter, which decomposes one verdict into eight teachable reasons and applies progressive disclosure. The result column is placed at the right of each row where colour and text are read together, and the relationship scatter is positioned beneath the table so the lesson moves from individual gates to the overall tendency between setup strength and outcome.

wf

The Institutional dashboard wireframe leads with two rows of key performance indicator cards because a governance reader requires headline metrics readable in seconds, and the tolerance flag sits directly beneath them so a raw drawdown figure is converted into a within tolerance judgement at the point of reading. The per pair table provides the grouped breakdown, and the cumulative line and horizontal bar charts occupy the foot so the trend and the categorical comparison support rather than crowd the headline figures.

wf

The Operational dashboard wireframe places the system status table at the top with a traffic light in each row so an administrator scanning for problems reaches an instant operational, degraded or failed judgement. The evaluation cycle log sits below to hold the data behind the error rate, and the compact

error trend is positioned beside it so reliability over time is visible without breaking the one row per component layout that keeps the dashboard scannable.

2.10 Explanation of Trends, Patterns, Relationships and Insights Each Dashboard Reveals

Trader Dashboard Insights

The trader dashboard reveals signal consistency and drawdown recovery. Observing that recent comparable signals were profitable builds confidence in executing the current decision, while the equity curve reveals whether recent trades are recovering or extending losses.

Stated precisely, the trend the trader reads is the slope of the equity curve over recent trades, the pattern is the run of greens and reds in the recent-results column that signals a winning or losing streak, and the relationship is between the confidence score attached to a signal and the size of the risk-reward target the system offers for it. The insight that emerges from these together is whether the present moment is a favourable one to act on a signal: a rising curve, a recent run of wins, and a high-confidence signal align into a clear decision to take the trade, whereas a falling curve and a low-confidence signal counsel caution even when a signal is present.

Coach Dashboard Insights

The coach dashboard reveals filter-logic transparency and the relationship between setup quality and outcome. The filter table provides visual proof of why a setup was accepted or rejected, and the scatter plot reveals that stronger displacement tends toward profitability, supporting the teaching that setup quality matters.

Stated precisely, the pattern the coach draws out is which combinations of passing filters tend to precede profitable trades, the relationship is the upward tendency of profit as displacement strength increases, visible as a drift of points from lower-left to upper-right in the scatter plot, and the trend across many examples is that setups passing all eight filters succeed more often than those scraping through on the minimum. The insight is pedagogical: a student learns that the system's value lies not in any single filter but in the confluence of all of them, and that the strength of a setup, not merely its existence, is what makes it worth taking.

Institutional Dashboard Insights

The institutional dashboard reveals overall profitability and risk, period-comparison trends, and symbol-level variation. Comparing one period against another reveals whether recent performance is typical, and the symbol breakdown reveals which pairs warrant increased or reduced allocation.

Stated precisely, the trend is the direction of cumulative return across weeks and months, the pattern is the relative consistency of win rate from one currency pair to the next, And the relationship is between risk taken, read as maximum drawdown, and reward earned, read as return, which together express whether profit was won safely or through excessive exposure. The insight that supports the governance decision is whether the strategy deserves continued or increased capital: a steady upward trend, consistent win rates, and a drawdown within tolerance justify allocation, while strong returns accompanied by a deep drawdown flag a strategy that is profitable but risky and may warrant reduced sizing.

Operational Dashboard Insights

The operational dashboard reveals system soundness and trend information. All-green status confirms results reflect true strategy performance, while a flashing error rate across successive evaluation cycles reveals a developing problem. Because the data is static, these insights describe the integrity of the stored dataset and the evaluation process, not the liveness of a feed.

2.11 Description of Accessibility and Inclusive Design Considerations

Colour Reliance

Because the dashboards depend on a green and red convention that disadvantages users with red-green colour blindness, every colour-coded cell is supplemented with a text label, so PASS and FAIL, or WIN and LOSS, are conveyed by word as well as by colour. Charts carry data labels so meaning never rests on colour alone.

Visual Accessibility

Labels use eleven-point or larger fonts and key indicators use eighteen-point or larger, and the worksheets remain legible under application zoom, with charts resizing to match.

Motor and Non-Mouse Access

The dashboards are fully keyboard-navigable, requiring no drag-and-drop, so users with motor impairments can move between cells, follow hyperlinks, and read cell comments without a mouse.

Screen Reader Compatibility

Critical information is presented in compact summary tables rather than long rows so that screen readers such as JAWS and NVDA can convey the headline data without traversing hundreds of cells, with the institutional summary placed at the top of its worksheet for this reason.

Stage 3:

This stage produces the data visualisation solution in Microsoft Excel and records, explains and analyses how each dashboard was implemented. The workbook builds the four stakeholder dashboards planned in Stage 2 on top of the existing Forex Decision Support System, with every figure derived from one shared dataset through a transparent formula chain. The development process, the implementation techniques and the way each dashboard communicates its trends, patterns and insights are documented section by section, and the live build and evaluation are demonstrated in the accompanying screen recording.

3.1 Description of the Implementation Approach and Workbook Architecture

The solution is implemented as a single Excel workbook of ten worksheets that deliberately separates the source data, the manual input point, the evaluation engine and the four stakeholder dashboards into distinct layers. This separation keeps one authoritative dataset feeding every view, which is the consistency principle established in Stage 2 and the single most important architectural decision in the build.

The worksheets fall into four layers. The data layer is the Historical_Data sheet, a static set of 150 EUR/USD bars that acts as the single source of truth. The control layer is the Input sheet, the only place a user types, together with the Logic sheet, which holds the eight filter gates and the displacement, confidence and risk calculations. The communication layer is the four dashboards, being Trader_Dashboard, Coach_Dashboard, Institutional_Dashboard and Operational_Dashboard. A supporting layer holds Calculation_Breakdown, a teaching view that exposes the derivation step by step, and Trade_History, the accumulating log that drives the portfolio aggregates. The tabs are ordered so a viewer meets the dashboards first and the engineering behind them second, which carries the progressive disclosure principle from the plan into the structure of the file itself.

Because the layers are connected by cell references rather than copied values, a change entered once on the Input sheet flows through the Logic engine and surfaces simultaneously across every dependent dashboard. No figure is ever typed twice and no dashboard holds a private copy of the data, which removes the most common source of inconsistency in spreadsheet reporting and guarantees that the four stakeholder views can never disagree with one another.

3.2 Description of the Historical Dataset Construction

The Historical_Data sheet was built first because every downstream calculation depends on it. The raw open, high, low and close values for 150 EUR/USD bars were imported as the source of truth, and a set of analytics columns was then layered on top to supply the inputs that the Decision Support System requires.

Every analytics column is produced by an Excel formula rather than a typed number, so the dataset recalculates in full if the underlying prices are ever replaced. Body is the absolute difference between open and close, Range is high minus low, and Efficiency is body divided by range. True Range and a fourteen period ATR are computed for the volatility and risk rules. The twenty one period EMA is built with a recursive formula that weights each close at two divided by twenty two and carries the previous EMA forward, which reproduces the higher timeframe trend reference the system relies on. AvgBody14, RecentRange5, PriorRange5 and a RegimeRatio complete the set so the eight filters have every value they need.

The sheet is hidden and treated as read only. Hiding it keeps the source of truth out of reach of accidental edits while leaving it fully available to the formulas that reference it, which is the first of several data integrity measures described in section 3.9.

3.3 Explanation of the Evaluation Engine Implementation

The Logic sheet is the engine that converts a new bar into a decision, and it was implemented to reproduce the exact rules of the Phase 1 system so that the dashboards report the real behaviour of the Decision Support System rather than a simplified imitation.

The engine first computes the displacement ratio as the candle body divided by the fourteen bar average body, then converts it to a confidence score using the rule that confidence equals the displacement ratio minus one, divided by two, held between zero and one. The confidence score selects the risk to reward tier, where a score at or above 0.85 targets three to one, a score at or above 0.5 targets two to one, and anything lower targets one to one. The eight Module E filters then run as binary gates, each implemented with an IF function that shows its tested value beside a PASS or FAIL result. The gates cover the expanding regime check, the volatility band, retest efficiency, higher timeframe trend alignment, displacement strength, fair value gap size, a clean retest close, and the spacing and lockout check. The final decision is a single IF wrapped around an AND of all eight gates, which returns TAKE only when every gate passes and REJECT otherwise, faithful to the system rule that no partial credit is awarded.

The risk parameters are placed by the same formula chain. The stop loss buffer scales with confidence, the risk distance is held to a minimum of two times ATR, and the take profit is set at the tier multiplied by the risk distance and capped at six times ATR. With the bar currently loaded on the Input sheet the engine returns TAKE for a BUY at a confidence of 1.0 with all eight gates passing, an entry of 1.0847, a stop at 1.07835 and a target at 1.09554, which demonstrates the complete pipeline producing a live and internally consistent result that the screen recording walks through end to end.

3.4 Implementation and Analysis of the Trader Dashboard

Development process and features

The Trader_Dashboard was built to answer a single question for the primary end user in seconds, which is whether to take the current setup. The development process began by placing the engine decision as a large headline cell, then adding the six parameters a trader acts on, being direction, confidence, risk to reward, entry, stop and target, as linked cells beside it. A recent signals table was then added beneath, drawing the last ten rows from Trade_History, and two charts were placed below that, a column chart of recent profit and loss and a line chart of the cumulative equity curve.

Implementation techniques

The implementation centres on linked cells and conditional formatting. The headline reads the engine result directly so the dashboard and the engine can never disagree, and a conditional formatting rule fills it green for TAKE and red for REJECT so the answer is legible before any number is read. The recent signals table uses conditional formatting to shade wins green and losses red and to colour each profit and loss value by its sign. The column chart is configured to plot trade outcomes as discrete bars and the line chart to plot the running pip total as a continuous series, which are the chart types justified for this view in Stage 2.

Communication of trends, patterns and insights

This view communicates an insight, a trend and a pattern at once. The insight is the take or reject decision carried by the headline, the trend is the slope of the rising equity curve, and the pattern is the recent run of wins and losses in the table. The visual hierarchy places the single most important output at the top left where the eye lands first, which is exactly the behaviour the time pressured trader persona needs, and the effect is a dashboard that delivers a confident decision and the recent context for it without the trader having to interpret a single raw calculation.